voltage of the transformer is $+E$, the potential of the point d, in a stationary state, fluctuates between 0 and $2E$, and the point c has the constant potential $2E$. No current flows then through the valves. Similarly, we find that in a stationary state the points e, g, i have the constant potentials $4E$, $6E$ and $8E$, while the potential of the points f, h, k fluctuates between $2E$ and $4E$, between $4E$ and $6E$, and between $6E$ and $8E$, respectively. When, for instance, a current enters at the point i, the potential there decreases slightly, and the valves let through a beam of electrons in the direction of the arrow; these electrons carry the charge along, so that the potential at the point i cannot drop very far below $8E$. Thus, proceeding by n steps, we obtain a $2n$-fold of the peak voltage of the transformer—for instance, when the initial voltage E is 200 kv and three steps are used, the direct-current voltage ultimately obtained is 1,200,000 volts.

Figure 34 shows an exterior view of the high-voltage generator of the *Kaiser Wilhelm Institut für Physik* in Berlin-Dahlem. The slanting parts are the valves, and the globes correspond to the points c, e, f in Figure 33.

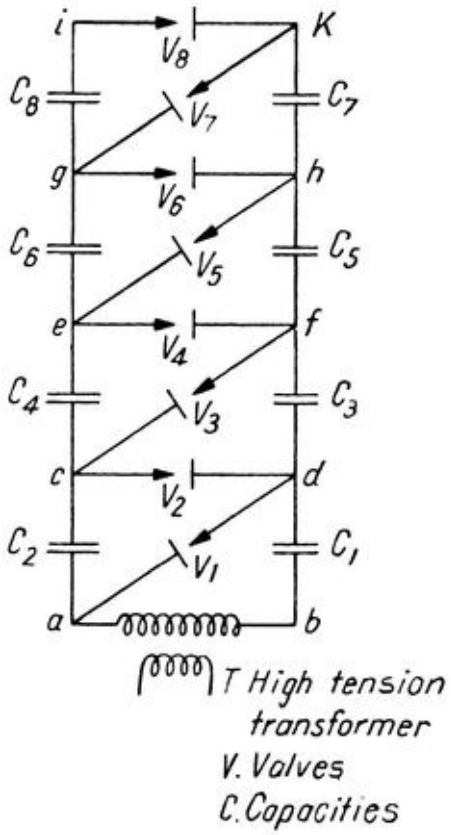


Figure 33.—Greinacher's form.

The high voltage thus produced must now be used to accelerate charged particles. These latter originate as canal rays in an ordinary discharge tube. They then enter the highly evacuated accelerator tube, with the high voltage drop between its terminals. At the end of the accelerator tube they hit the